

```
1  ../README.txt
2
3  favorites/favorites{0,1,2,3,4}.py
4  favorites/favorites{5,6,7,8}.py
5  favorites/favorites{9,10}.py
```

```
1  # Prints all favorites in CSV using csv.reader
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create reader
9      reader = csv.reader(file)
10
11     # Skip header row
12     next(reader)
13
14     # Iterate over CSV file, printing each favorite
15     for row in reader:
16         print(row[1])
```

```
1  # Stores favorite in a variable
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create reader
9      reader = csv.reader(file)
10
11     # Skip header row
12     next(reader)
13
14     # Iterate over CSV file, printing each favorite
15     for row in reader:
16         favorite = row[1]
17         print(favorite)
```

```
1  # Prints all favorites in CSV using csv.DictReader
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Iterate over CSV file, printing each favorite
12     for row in reader:
13         favorite = row["language"]
14         print(favorite)
```

```
1  # Prints all favorites in CSV using csv.DictReader
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Iterate over CSV file, printing each favorite
12     for row in reader:
13         print(row["language"])
```

```
1  # Counts favorites using variables
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Counts
12     scratch, c, python = 0, 0, 0
13
14     # Iterate over CSV file, counting favorites
15     for row in reader:
16         favorite = row["language"]
17         if favorite == "Scratch":
18             scratch += 1
19         elif favorite == "C":
20             c += 1
21         elif favorite == "Python":
22             python += 1
23
24     # Print counts
25     print(f"Scratch: {scratch}")
26     print(f"C: {c}")
27     print(f"Python: {python}")
```

```
1  # Counts favorites using dictionary
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Counts
12     counts = {}
13
14     # Iterate over CSV file, counting favorites
15     for row in reader:
16         favorite = row["language"]
17         if favorite in counts:
18             counts[favorite] += 1
19         else:
20             counts[favorite] = 1
21
22     # Print counts
23     for favorite in counts:
24         print(f"{favorite}: {counts[favorite]}")
```

```
1  # Uses try/except instead
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Counts
12     counts = {}
13
14     # Iterate over CSV file, counting favorites
15     for row in reader:
16         favorite = row["language"]
17         try:
18             counts[favorite] += 1
19         except KeyError:
20             counts[favorite] = 1
21
22     # Print counts
23     for favorite in counts:
24         print(f"{favorite}: {counts[favorite]}")
```



```
1  # Sorts favorites by key
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Counts
12     counts = {}
13
14     # Iterate over CSV file, counting favorites
15     for row in reader:
16         favorite = row["language"]
17         if favorite in counts:
18             counts[favorite] += 1
19         else:
20             counts[favorite] = 1
21
22     # Print counts
23     for favorite in sorted(counts):
24         print(f"{favorite}: {counts[favorite]}")
```

```
1  # Sorts favorites by value using .get
2
3  import csv
4
5  # Open CSV file
6  with open("favorites.csv", "r") as file:
7
8      # Create DictReader
9      reader = csv.DictReader(file)
10
11     # Counts
12     counts = {}
13
14     # Iterate over CSV file, counting favorites
15     for row in reader:
16         favorite = row["language"]
17         if favorite in counts:
18             counts[favorite] += 1
19         else:
20             counts[favorite] = 1
21
22     # Print counts
23     for favorite in sorted(counts, key=counts.get, reverse=True):
24         print(f"{favorite}: {counts[favorite]}")
```

```
1  # Sorts favorites by value using SQL
2
3  from cs50 import SQL
4
5  # Open database
6  db = SQL("sqlite:///favorites.db")
7
8  # Search for title
9  rows = db.execute("SELECT language, COUNT(*) AS n FROM favorites GROUP BY language ORDER BY n DESC")
10
11 # Print popularity
12 for row in rows:
13     print(row["language"], row["n"])
```

```
1  # Searches database for popularity of a problem
2
3  from cs50 import SQL
4
5  # Open database
6  db = SQL("sqlite:///favorites.db")
7
8  # Prompt user for favorite
9  favorite = input("Favorite: ")
10
11 # Search for title
12 rows = db.execute("SELECT COUNT(*) AS n FROM favorites WHERE problem = ?", favorite)
13
14 # Get first (and only) row
15 row = rows[0]
16
17 # Print popularity
18 print(row["n"])
```